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Thinking in the Rectified Latent Manifold: Dual-Anchored Alignment for Robust Remote Sensing Visual Grounding

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Thinking in the Rectified Latent Manifold: Dual-Anchored Alignment for Robust Remote Sensing Visual Grounding

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Abstract—Visual grounding in remote sensing (RS) aims to localize target objects specified by natural language instructions. Despite the impressive success of existing task-specific discriminative models and generative multimodal Large Language Models (MLLMs) under idealized environments, their performance sharply degrades when confronted with real-world joint text-image perturbations, such as atmospheric corruptions and linguistic noise. In this paper, we investigate the root cause of this vulnerability and reveal that joint perturbations trigger severe high-dimensional feature shifts within the hidden-state space, which ultimately disrupts the delicate text-visual semantic alignment. To address this fundamental limitation, we propose interleaved text-image chains, a novel multimodal reasoning paradigm that represents image sequences via latent semantic expressions to elegantly unify low-level image enhancement and high-level image understanding within a cohesive reasoning trajectory. Based on this paradigm, we introduce the dual-anchored manifold alignment (DAMA) framework. By incorporating latent visual alignment (LVA) and textual space alignment (TSA), DAMA explicitly projects shifted, corrupted feature distributions back onto a stable, rectified dual-anchored latent manifold (DALM), effectively insulating downstream semantic processing from upstream disturbances. We optimize DAMA via a two-stage training paradigm: dual-anchored supervised fine-tuning for foundational reasoning, followed by outcome-driven reinforcement learning via LE-GRPO, which leverages the intermediate thinking process as a flexible workspace for unconstrained latent exploration and autonomous alignment correction. Extensive experiments on the constructed benchmark characterized by joint text-image perturbations demonstrate that our framework consistently achieves state-of-the-art (SOTA) performance.

Index Terms—Visual Grounding, remote sensing images, multimodal large language models (MLLMs), latent manifold alignment.

I. INTRODUCTION

Visual grounding in remote sensing (RS) aims to localize target objects specified by natural language instructions. This task holds immense potential for real-world applications such as disaster monitoring, military surveillance, agricultural management, and emergency response [1]–[4]. Recently,

an increasing number of transformer-based approaches and multimodal large language models (MLLMs) have achieved impressive performance under standard benchmark environments [5], [6]. However, in practical deployment scenarios, the inputs frequently suffer from complex degradations rather than adhering to idealized clean benchmarks [7]–[10]. Specifically, on the visual side, optical RS images are routinely corrupted by atmospheric interference, such as clouds and fog [11], [12], as well as sensor-induced degradations like noise [13] and motion blur [14]. Concurrently, on the textual side, human-generated instructions naturally introduce linguistic noise, including colloquial expressions, verbose descriptions, and typographical errors [15]–[17]. Consequently, there is an **urgent need** for robust models capable of yielding accurate and consistent predictions, maintaining high fidelity under both idealized conditions and the severely degraded text-image queries typical of actual operations.

Driven by the rapid advancement of deep learning (DL) algorithms, the landscape of visual grounding has undergone continuous and profound evolution, which can be broadly categorized into several historical stages. In the pioneering era prior to 2014, formal visual grounding frameworks were not yet fully established, instead, the task primarily served as a downstream validation metric for referring expression generation (REG). From 2014 to 2020, the paradigm shifted toward joint feature encoding, typically employing small-scale long short-term memory (LSTM) networks [18] for textual modeling alongside convolutional neural networks (CNNs) [19] for visual feature extraction. Subsequently, around 2021, a major architectural transition occurred as LSTMs and CNNs were systematically replaced by the transformer architecture [20], which quickly popularized the "pre-training then fine-tuning" paradigm for downstream tasks. During this era, pre-trained language encoders (e.g., BERT [21]), object detection frameworks (e.g., DETR [22]), and foundation vision-language models (VLMs) (e.g., CLIP [23] and BEiT-3 [24]) were extensively adapted to visual grounding, typically optimizing discriminative alignment scores or bounding box regression heads. More recently, fueled by exponential growth in computational power, massive training datasets, and the empirical validation of scaling laws [25], the field since 2023 has forcefully entered the era of Generative MLLMs [26], where these massive foundational targets are instruction-tuned to autoregressively predict target coordinates [27]. Currently, state-of-the-art research predominantly converges on these last two

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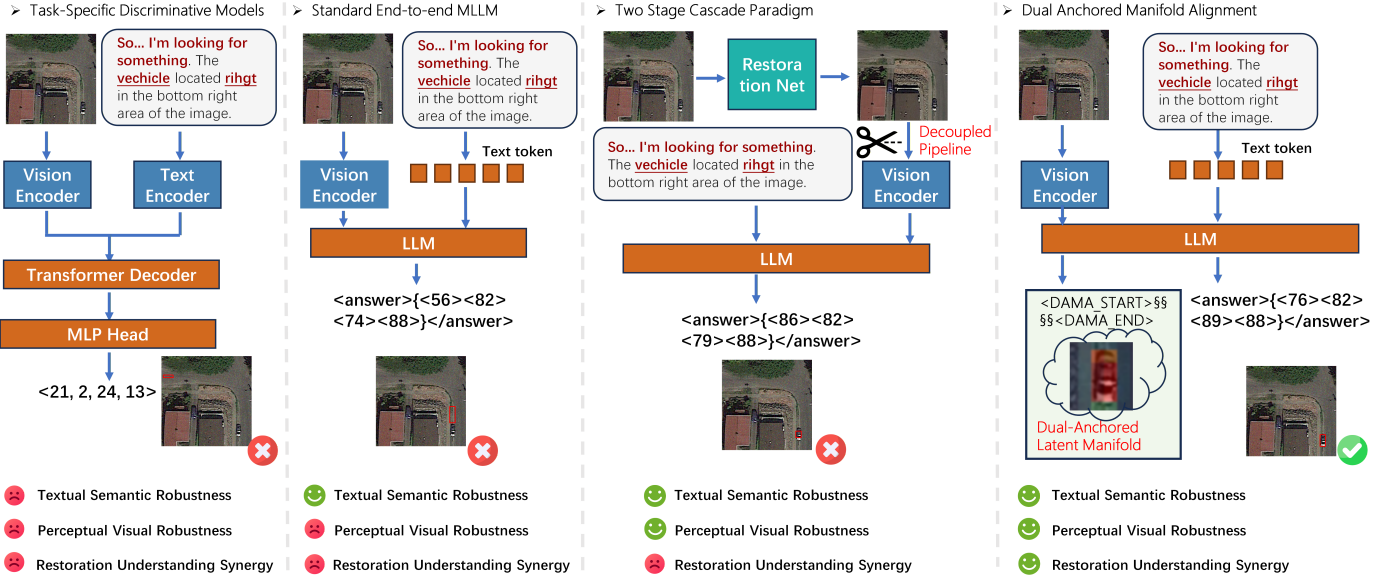


Fig. 1: Comparison of different architectural paradigms for visual grounding under joint text and image perturbations. **(a) Task-specific discriminative paradigm.** Approaches relying on rigid text and visual encoders exhibit extreme vulnerability under colloquial text variations and degraded input images due to their fragile, unrectified representation spaces. **(b) Standard end-to-end MLLMs.** These models exhibit partial textual semantic robustness owing to the context-aware reasoning capabilities of LLMs, however, concurrent image degradations trigger severe high-dimensional feature drift within the visual space, leading to poor perceptual visual robustness. **(c) Two-stage cascade paradigm.** This paradigm suffers from a decoupled pipeline that lacks mutual synergy between image restoration and semantic understanding. Furthermore, blind image restoration often inadvertently weakens critical grounding cues by over-smoothing mission-critical target features. **(d) Dual anchored manifold alignment (Ours).** Our framework integrates a dual anchored manifold alignment mechanism directly within the multimodal stream. While joint perturbations cause severe feature dislocation, our approach explicitly projects the corrupted high-dimensional distributions back onto a stable, clean, and rectified latent manifold. Within this space, the Latent Visual Alignment module aligns internal hidden states with idealized target representations to reconstruct clean visual features, enabling the model to derive reliable predictions from the calibrated space, thereby achieving highly robust visual grounding performance

contemporary paradigms, which we define as **Task-Specific Discriminative Models** and **Generative MLLMs**.

Task-specific discriminative models and standard MLLMs exhibit superior performance and strong visual-semantic understanding on standard benchmarks [28], [29], with the latter even outputting fine-grained grounding answers under complex textual semantics by exploiting the representation power of LLMs. However, these approaches fail to maintain robustness when confronted with joint text and image perturbations. As illustrated in Fig. 1, the task-specific discriminative paradigm utilizes independent vision and text encoders to extract visual and textual features, fuses them via cross-attention mechanisms within a transformer decoder, and regresses the final coordinates through an MLP head. Nonetheless, its heavy reliance on rigid, frozen encoders pre-trained on pristine image-text pairs renders it highly vulnerable to degraded input images and non-canonical textual variations. Standard MLLMs, on the other hand, exhibit a degree of textual semantic robustness owing to the inherent reasoning capabilities of their LLM backbones, nevertheless, environmental image degradations trigger severe high-dimensional feature shifts within the visual representation space, and then the distribution shift disrupts the delicate text-visual semantic alignment, forcing the model to operate

on distorted feature representations and then output inaccurate grounding boundaries. Alternatively, the two-stage cascade paradigm attempts to decouple the problem by first deploying a restoration network to restore the corrupted images, and then employing an MLLM for the final visual grounding task. Although this paradigm alleviates the joint perturbation issue to some extent, it suffers from a fragmented pipeline that lacks mutual synergy between low-level restoration and high-level semantic understanding. Furthermore, blind image restoration often inadvertently **attenuates** critical grounding cues by over-smoothing mission-critical target features, yielding sub-optimal results. To circumvent these architectural limitations, alternative approaches resort to data augmentation by incorporating noisy image-text pairs during training. However, since synthesized training datasets can hardly simulate the full spectrum of complex, real-world environments, composite multi-modal perturbations remain inherently **insufficiently covered** within a long-tail distribution. Consequently, model performance exhibits extreme sensitivity during out-of-distribution (OOD) deployments, **thereby forcing** the visual encoder to struggle with unfamiliar degraded inputs and **ultimately propagating corrupted** visual semantics throughout the subsequent reasoning trajectory.

To empirically validate the aforementioned vulnerability,

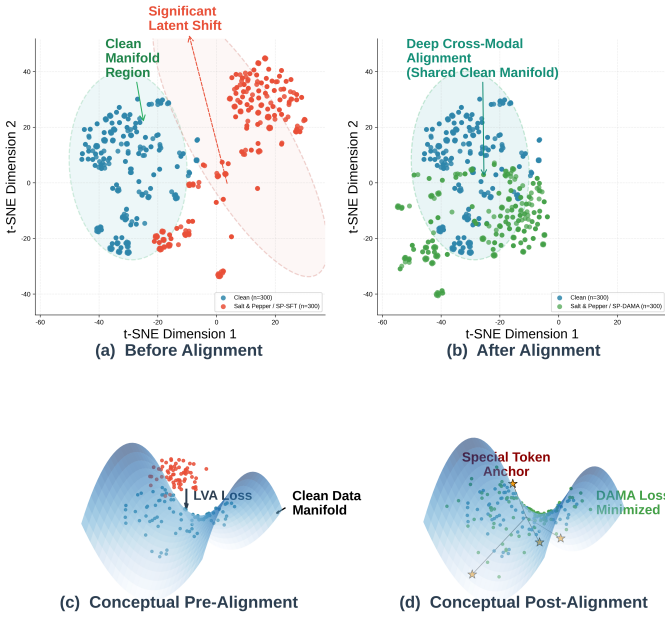


Fig. 2: Qualitative analysis of the intermediate latent space topology under multimodal perturbations, extracted from the 10-th layer hidden states ($n = 300$ samples per distribution). (a) **Before Alignment (t-SNE)**: Perturbations cause a severe latent shift, drifting corrupted features (red dots) away from the clean distribution (blue dots). (b) **After Alignment (t-SNE)**: DAMA achieves deep cross-modal alignment, pulling corrupted embeddings (green dots) back into the shared clean manifold. (c) **Conceptual Pre-Alignment**: Visual noise detaches features from the continuous clean manifold, contaminating the latent topology and yielding a high LVA loss. (d) **Conceptual Post-Alignment**: The LVA constraint forces features back onto the manifold surface, while Special Token Anchors secure textual boundaries for dual-anchored correction.

we construct a perturbed evaluation dataset comprising both image-level degradations [30] and textual perturbations [9]. We then evaluate state-of-the-art MLLMs using these perturbed samples and perform an in-depth analysis of their hidden-state spaces and distributions. Through our preliminary exploration (taking a fine-tuned Qwen-VL model as a representative case), we observe a severe performance degradation under joint text and image perturbations. Investigating the root cause, we discover that the corrupted hidden states exhibit a significant distributional shift away from the clean reference distribution, thereby distorting the extracted visual representations in the latent space and disrupting the delicate text-visual semantic alignment, as shown in Fig. 2.

Motivated by these observations, we propose a novel **dual-anchored manifold alignment (DAMA)** framework. DAMA explicitly projects shifted, corrupted high-dimensional features back onto a stable, clean representation space, termed the dual-anchored latent manifold (DALM). Specifically, **DAMA generates mental imagery within the latent space before deriving the final bounding box**, driven by two synergistic components: latent visual alignment (LVA) and textual space alignment (TSA). LVA aligns the MLLM’s hidden states with

idealized target representations to **reconstruct clean visual features**, effectively rectifying distorted trajectories into a pristine visual-semantic manifold that precisely localizes the target object. Concurrently, TSA regularizes these latent embeddings by anchoring them to designated tokens within the textual space, thereby preventing the reasoning path from **diverging into** incoherent latent states or inducing hallucinations during subsequent autoregressive generation. By confining the reasoning trajectory within this rectified latent manifold, our framework effectively shields downstream semantic decoding from upstream multi-modal perturbations, thereby ensuring robust and precise visual grounding. In summary, the core contributions of this work are threefold,

- We propose **interleaved text-image chains**, a novel multimodal reasoning paradigm that represents image sequences via latent semantic expressions, thereby elegantly unifying low-level image enhancement and high-level image understanding within a single, cohesive reasoning trajectory.
- We introduce the **dual-anchored manifold alignment (DAMA)** framework, which explicitly projects corrupted, shifted high-dimensional hidden states back onto a stable and rectified latent manifold. This mechanism successfully insulates downstream multimodal reasoning from upstream environmental and linguistic disturbances, thereby ensuring a highly robust and precise semantic reasoning trajectory.
- We develop a comprehensive two-stage training paradigm to optimize our framework. It begins with **dual-anchored supervised fine-tuning** to endow the model with fundamental visual reasoning capabilities, and advances to **outcome-driven reinforcement learning via Lanet exploration GRPO** to stimulate latent reasoning and rectify the trajectory. By treating the intermediate thinking process as a flexible workspace, this reinforcement stage allows the model to undergo unconstrained latent exploration to autonomously correct its alignment path. Consequently, extensive experiments on the two benchmark characterized by severe joint image and text perturbations demonstrate that our approach consistently achieves state-of-the-art (SOTA) performance.

II. RELATED WORK

A. Visual Grounding and its application in RS domain

Visual grounding aims to localize target object regions specified by natural language instructions. While initially developed for natural images, this task has recently been extended to the RS domain. In natural imagery, visual grounding methodologies can be broadly categorized into four paradigms: two-stage, one-stage, transformer-based, and MLLM-based approaches.

Specifically, two-stage methods first utilize a pre-trained detector to generate a pool of candidate regions, and subsequently select the best-matching region that aligns with the linguistic query [31]–[35]. For instance, the dynamic graph attention (DGA) network [31] constructs image relational graphs

and employs a differential analyzer to perform language-guided, multi-step reasoning over visual components. However, the efficacy of these two-stage approaches is heavily dependent on the quality of the frozen object detectors. Moreover, the decoupled, two-stage pipeline inherently bottlenecks inference efficiency.

In contrast to two-stage paradigms, one-stage approaches eliminate the decoupled region proposal step by directly integrating multimodal features to predict object bounding boxes in an end-to-end manner [36]–[41]. For instance, FAOA [36] leverages the visual backbone of YOLOv3 and fuses textual embeddings to perform direct, single-shot bounding box regression. Shifting away from conventional anchor-based regression, SeqTR [41] reformulates visual grounding as a point-prediction task, framing it as a sequence-oriented representation problem where bounding boxes are tokenized into discrete coordinate sequences. Concurrently, the rapid ascent of transformers in computer vision has catalyzed a paradigm shift in visual grounding architectures [22], [42]–[45]. As a pioneer in this line of work, TransVG [42] utilizes a transformer encoder to visually and textually fuse cross-modal features, treating grounding as a direct coordinate regression task. To further optimize inference efficiency, its successor, TransVG++ [43], discards the independent fusion module entirely. Instead, it injects textual information directly into the intermediate layers of the vision backbone, thereby achieving a more streamlined and computationally efficient architecture.

Driven by the scaling laws of LLMs, MLLMs have increasingly been leveraged to revolutionize visual grounding tasks [26], [27], [46], [47]. Beyond global image-text comprehension, specifically, to achieve fine-grained grounding, Ferret [27] unifies referring and localization via a hybrid region representation that combines discrete coordinates and continuous features, accommodating spatial queries of arbitrary shapes. Addressing the granularity limitations of coarsely pre-trained vision backbones, LION [47] injects dual-level visual knowledge through a mixture-of-adapters architecture and soft-prompted semantic tags. Extending this paradigm to dynamic environments, VideoGLaMM [48] tackles complex spatio-temporal dynamics by bridging a dual vision encoder and a mask decoder via bidirectional adapters for pixel-level video tracking and grounding.

Compared to natural images, visual grounding in RS scenarios introduces distinct challenges [49], [50]. First, target objects in RS imagery typically exhibit extreme scale variations and are embedded in highly cluttered, complex backgrounds. Second, the multi-scale visual features of different candidate objects frequently display a high degree of semantic similarity, exacerbating cross-modal ambiguity. To address these challenges, GeoVG [49] pioneered the RS visual grounding task by curating a benchmark dataset and constructing a spatial relational graph that integrates textual geographic relations with visual features via attention mechanisms. More recently, inspired by breakthroughs in natural language processing, several frameworks have integrated VLMs and LLMs to exploit their advanced reasoning capabilities for RS visual grounding [51]–[53]. For instance, Geo-CoT incorporates explicit geospatial reasoning into the grounding process, while SegEarth-

R1 leverages a large-scale multimodal architecture to learn a unified language-guided grounding and segmentation pipeline. Despite their architectural advancements, these approaches demonstrate fragile robustness and suffer severe performance degradation when confronted with joint textual and visual perturbations.

B. Robustness of Multimodal Models

Enhancing the robustness of MLLMs is crucial as the evaluation paradigm transitions from clean academic benchmarks to unpredictable, real-world deployment scenarios. Recent literature reveals that existing VLMs and MLLMs remain vulnerable to out-of-distribution shifts and environmental perturbations, consequently, models that excel in standard laboratory settings often experience a sharp degradation in performance when subjected to multimodal perturbations in realistic scenarios [54]–[56].

To systematically diagnose this fragility, current benchmarking studies generally evaluate model robustness through two aspects: visual and textual dimensions. On the visual side, image perturbations can severely distort the feature distributions output by the vision encoder, thereby undermining the downstream multimodal reasoning process. These visual degradations encompass a diverse spectrum of adverse conditions, including atmospheric noise, motion blur, cloud obscuration, and complex composite mixtures of these factors [57], [58]. On the textual side, the text encoder frequently exhibit severe prompt sensitivity, where minor linguistic variations or textual perturbations can trigger significant reasoning failures [16]. To cultivate model invariance against such semantic shifts, recent studies have explored preference-driven optimization strategies. For instance, RobustFlow [59] mitigates the brittleness of LLM-generated agentic workflows by leveraging preference optimization over sets of synonymous task descriptions, thereby enforcing topological consistency under paraphrasing and noise injection. More specific to RS, RemoteShield [9] counters multimodal noise by organizing clean and co-corrupted inputs into semantic equivalence clusters, optimizing the MLLM via cross-condition preference learning. However, although such preference learning forces consistent final predictions, it does not explicitly isolate or rectify the degradation information embedded within the latent features, leaving the internal representations contaminated by multimodal corruptions.

III. METHOD

A. Overview

To overcome representation drift and performance degradation induced by cross-modal perturbations, ranging from visual distortions (e.g. blur, compression, and noise) to linguistic variations (e.g. naturalistic and persona driven paraphrasing), we propose **Dual-Anchored Manifold Alignment (DAMA)** approach, a latent visual reasoning framework specifically designed for robust visual grounding, as shown in Fig. 3. Inspired by human cognitive paradigms, DAMA generates mental imagery within a latent space before deriving the final bounding box. Specifically, we introduce **DAMA** to

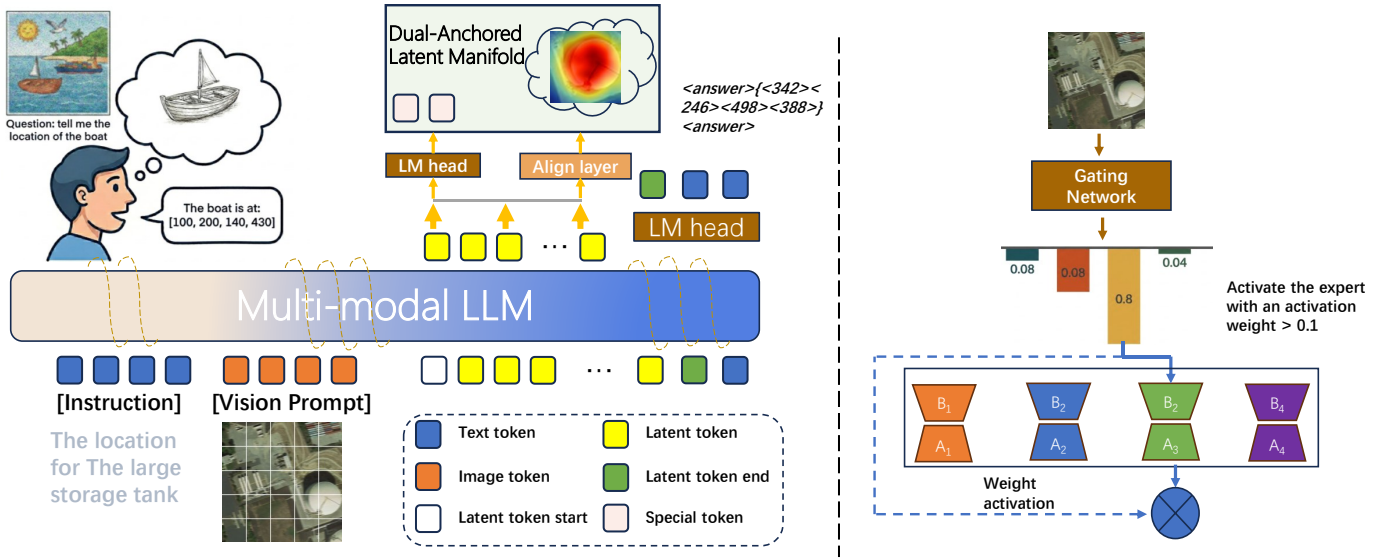


Fig. 3: An overview of the DAMA framework. **Left:** For visual grounding tasks under degraded image inputs, the MLLM executes a two-stage paradigm consisting of latent reasoning and final text generation, wherein DAMA generates mental imagery within a latent space prior to deriving the final bounding box. Because visual and text perturbations cause features to drift within the high dimensional space and deviate from the optimal clean distribution, we constrain the intermediate workspace within a **dual-anchored latent manifold** to reinforce model robustness. Specifically, the latent trajectory is regularized by simultaneously aligning with the visual semantics of the target object and anchoring to special tokens in the text space.

govern this latent workspace, which constrains the reasoning trajectory by anchoring it to both a clean visual manifold and a bounded semantic space.

In the following sections, we first detail the construction of a robust visual grounding dataset. Subsequently, we present a two-stage training pipeline, **dual-anchored supervised fine-tuning (DA-SFT)** and **latent exploration-constrained GRPO (LE-GRPO)**, both operating within the proposed latent space. Finally, to handle diverse degradation patterns, we incorporate a Hybrid Multi-LoRA Architecture, which effectively decouples representations to prevent interference across varying image perturbation types.

B. Robust Multimodal Perturbation and Verification Framework

Existing MLLMs for RS demonstrate satisfactory performance on standard benchmarks containing clean text and images. However, they typically fail to generalize to noisy conditions encountered in practical applications, where textual instructions and collected image samples are frequently susceptible to diverse environmental and human disturbances. To bolster model robustness and improve RS interpretation accuracy under these challenging scenarios, as shown in Fig. 4, we construct a comprehensive multimodal degradation dataset. This dataset consists of textual perturbations that mimic realistic human query variations alongside synthetic image perturbations that simulate practical environmental degradation effects. These visual corruptions encompass sensor acquisition degradations, atmospheric scattering, and digital transmission artifacts. Subsequently, we utilize an advanced VLM as a

quality control filter to eliminate unreasonable or semantically mismatched image text pairs.

Text Perturbation. On the textual side, we aim to simulate human text query variations in real scenes while preserving the original core semantics. In practical scenarios, human queries are often redundant, colloquial, and conversational, even when the underlying intent remains unchanged. To reproduce this variability, we leverage **Qwen3-14B** [60] to rewrite each query, altering the surface structure while anchoring the essential semantics. We establish two representative categories of human-centric text variations:

- **Naturalistic Style:** It incorporates highly colloquial expressions and conversational fillers such as "okay so", "like", and "so" to mimic spontaneous human speech. It occasionally introduces minor spelling errors, shorthand notations (such as "u", "pls", or "rn"), and mild linguistic noise to replicate real world typing habits and immediate self corrections.
- **Persona Style:** It features context dependent reformulations shaped by urgency, fluctuating emotional tones, and reassurance loops that demand explicit reconfirmation.

Image Perturbation. We simulate four representative image perturbations categorized into three distinct domains: sensor induced degradations, atmospheric scattering, and digital transmission degradations.

- **Sensor Induced Degradations:** It introduces *salt-and-pepper noise* [61] to simulate bit-flip errors, which drastically reduce classification and interpretation accuracy. Furthermore, it incorporates *motion blur* [62] arising from

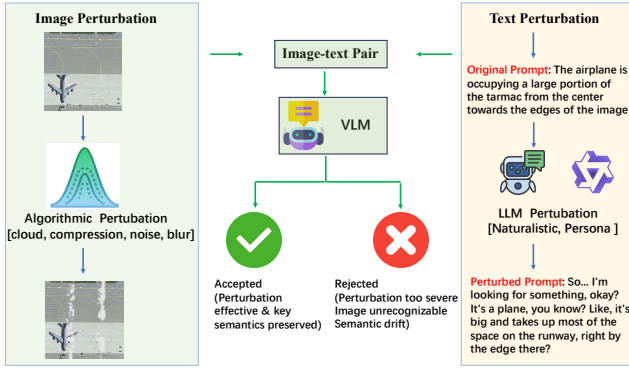


Fig. 4: Overview of our dataset construction pipeline. We first generate text perturbations and image perturbations using LLMs and analytical simulation algorithms. Subsequently, we utilize a VLM to exclude overly severe perturbations or cases where unrecognizable images cause semantic drift.

platform vibrations or rapid sensor movement, both of which degrade object detection and tracking performance.

- **Atmospheric Scattering Degradations:** *Cloud cover* [12], [63] occasionally occurs in optical RS data, severely hindering scene interpretation and object visibility.
- **Digital Transmission Degradations:** It incorporates *compression artifacts* [64], such as JPEG or JPEG 2000, which distort the high level features extracted by deep networks like CNNs or ViTs.

Since LLM rewriting is inherently stochastic, and certain image perturbations (such as heavy cloud cover) can completely obscure small target objects, we utilize an advanced VLM, **Qwen3-VL-32B**, to filter out invalid or unreasonable examples. We construct corresponding visual and text perturbation pairs to evaluate model robustness. However, during the two stage training process, we utilize standard textual instructions to ensure a fair comparison with baseline methods.

Each instruction tuning instance consists of an image question pair augmented with the bounding box coordinates of the target object. During the forward pass, the clean image is encoded by a ViT encoder to obtain raw visual embeddings $\mathbf{v} = \{\mathbf{v}_1, \dots, \mathbf{v}_{T_v}\}$. To extract precise visual semantics, we apply ROI Align to the feature map based on the target bounding box coordinates. This operation utilizes bilinear interpolation to extract precise, fixed-size spatial features while eliminating the quantization errors typical of standard ROI pooling. This yields a specific subset of target embeddings $\mathbf{v}_t = \{\mathbf{h}_1, \dots, \mathbf{h}_K\}$ (representing objects such as ships or storage tanks) that serve as ground truth visual anchors in the latent space, effectively supervising the alignment of the reasoning trajectory via our proposed DAMA mechanism.

C. Two-stage Training Pipeline

1) *Dual-Anchored Supervised Fine-Tuning:* To empower the MLLM with the capability to generate coherent and adaptive intermediate visual reasoning steps, we employ DA-SFT to instill the fundamental patterns of latent visual reasoning. During this stage, we supervise the latent reasoning

content via **dual anchored manifold alignment (DAMA)**. This mechanism forces the MLLM to utilize its latent space embeddings (e.g. the last hidden states) to reconstruct the clean visual features of the target object. Furthermore, to prevent the model from drifting into incoherent latent states or generating hallucinations during the reasoning process, we regularize these latent embeddings by anchoring them to special tokens within the text space. The auto-regressive latent reasoning format is formulated as,

$$\mathcal{S} = [\mathcal{X}, \langle \text{DAMA_START} \rangle, \mathbf{h}_1, \dots, \mathbf{h}_K, \langle \text{DAMA_END} \rangle, \mathcal{A}] \quad (1)$$

where $\mathcal{X}$ denotes the multimodal input, and $\mathcal{A}$ is the final text answer. $\langle \text{DAMA_START} \rangle$ and $\langle \text{DAMA_END} \rangle$ act as special markers to delineate the **Dual Anchored Latent Manifold (DALM)** interval. The sequence $\{\mathbf{h}_t\}_{t=1}^K$ represents the latent embeddings expected to encode the visual semantics of the target object within the aforementioned latent manifold, trained in a teacher-forcing manner.

The optimization of the proposed DAMA framework consists of two sequential objectives: the latent workspace alignment phase and the subsequent language modeling phase. We first introduce the DAMA loss, which serves to bind and regulate the latent reasoning trajectory within the $\langle \text{DAMA_START} \rangle$ and $\langle \text{DAMA_END} \rangle$ tokens through cooperative visual and textual constraints.

Latent Visual Alignment (LVA). To ground the intermediate thinking steps in physical reality, we extract the sequence of hidden states $\mathbf{h}_t$ generated within the latent reasoning interval. These hidden states are mapped into the visual semantics space via an alignment projector $\mathcal{P}_\theta$ consisting of a linear projection, layer normalization, and an activation function. The LVA loss minimizes the distance to the clean visual manifold,

$$\mathcal{L}_{\text{LVA}} = \frac{1}{K} \sum_{t=1}^K \|\mathcal{P}_\theta(\mathbf{h}_t) - \mathbf{v}_t\|_2^2 \quad (2)$$

Textual Space Alignment (TSA). Concurrently, to prevent the hidden states from drifting into an incoherent latent space that could degrade decoding stability, we utilize a causal language modeling objective across the entire reasoning block, including its boundaries. Formally, let the entire training sequence be represented as $\mathcal{S} = \{s_1, s_2, \dots, s_M\}$. Assuming the special tokens $\langle \text{DAMA_START} \rangle$ and $\langle \text{DAMA_END} \rangle$ are located at indices I_{start} and I_{end} respectively, the latent reasoning interval is defined by the inclusive index set,

$$\mathcal{T}_{\text{latent}} = \{t \mid I_{\text{start}} \leq t \leq I_{\text{end}}\} \quad (3)$$

To explicitly bridge the latent workspace and the textual token space, each token s_t within this interval is decoded from its corresponding hidden state $\mathbf{h}_t$ via the language model head, denoted as f_ϕ . The conditional probability of generating the target token s_t given the preceding context is formulated as,

$$P(s_t \mid s_{<t}) = \text{Softmax}(f_\phi(\mathbf{h}_t))[s_t] \quad (4)$$

where $[s_t]$ denotes the probability indexing for the specific ground truth token in the vocabulary. By executing next token prediction over this complete set, the TSA loss optimizes

the generation of the start marker, the intermediate thinking special tokens, and the end marker in a unified manner. This implicitly forces the entire latent trajectory to retain valid linguistic structures, formulated via cross entropy as:

$$\mathcal{L}_{\text{TSA}} = - \sum_{t \in \mathcal{T}_{\text{latent}}} \log P(s_t | s_{<t}) \quad (5)$$

where $s_{<t}$ denotes all preceding tokens in the sequence, encompassing the multimodal prompts and historical trajectory. The overall DAMA alignment objective is a joint optimization of both semantic boundaries:

$$\mathcal{L}_{\text{DAMA}} = \mathcal{L}_{\text{TSA}} + \lambda_{\text{LVA}} \mathcal{L}_{\text{LVA}}$$

where λ_{LVA} is a hyperparameter balancing the visual and textual alignment signals.

Next Token Prediction (NTP) Loss. Following the completion of the latent reasoning phase bounded by $\langle \text{DAMA_END} \rangle$, the model transitions to the language modeling phase to output the final grounding response. Conditioned on both the initial multimodal input $\mathcal{X}$ and the well aligned latent reasoning trajectory $\mathbf{h}_{1:K}$, the model generates the final response tokens $\{y_t\}_{t=1}^{T_y}$ to answer the visual question. We adopt the standard cross entropy loss to maximize the likelihood of the ground truth answer sequence,

$$\mathcal{L}_{\text{NTP}} = - \frac{1}{T_y} \sum_{t=1}^{T_y} \log p_{\theta}(y_t | y_{<t}, \mathbf{h}_{1:K}, \mathcal{X}). \quad (6)$$

During the supervised fine tuning stage, the total training loss is the combination of the latent workspace alignment and the final task completion,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DAMA}} + \mathcal{L}_{\text{NTP}} \quad (7)$$

2) Outcome-Driven Reinforcement Learning via LE-GRPO:

To unlock the full potential of adaptive visual reasoning and eliminate over-regulation within the latent workspace, we employ **Latent Exploration Group Relative Policy Optimization (LE-GRPO)** in the second training stage. Unlike the DA-SFT phase, we intentionally remove the explicit geometric visual alignment constraints ($\mathcal{L}_{\text{LVA}}$) from the latent trajectory, allowing the model to undergo unconstrained latent exploration within a flexible workspace.

However, extending the standard GRPO framework to govern the latent reasoning trajectory introduces a fundamental theoretical obstacle. Standard policy-gradient objectives are explicitly grounded in discrete action spaces governed by vocabulary distributions. In stark contrast, the internal reasoning activations within the DAMA workspace reside in a continuous, non-discrete feature manifold that completely lacks a well-defined token probability space, making conventional sequence-level alignment inapplicable.

To address this challenge, we introduce a teacher-forcing log-probability pass based on hidden-state replay. During the autoregressive rollout phase, for each multi-modal query $\mathcal{X}$, the old policy π_{ref} samples a group of G outputs and records the sequence of internal hidden states generated within the DAMA workspace, denoted as $\tilde{\mathbf{h}}_i = \{\mathbf{h}_{i,1}, \mathbf{h}_{i,2}, \dots, \mathbf{h}_{i,L}\}$. When evaluating the policy-gradient loss, rather than conditioning the importance ratio purely on textual history, these

recorded latent representations $\tilde{\mathbf{h}}_i$ are replayed and patch-injected into the attention layers as fixed Key-Value (KV) matrices under both π_{θ} and π_{ref} . This mechanism explicitly restores the exact continuous visual-semantic context preceding text generation. Accordingly, the token-level importance ratio $r_{i,t}(\theta)$ is conditioned directly on the replayed hidden states $\tilde{\mathbf{h}}_i$ along with the historical textual tokens $y_{i,<t}$:

$$r_{i,t}(\theta) = \frac{\pi_{\theta}(y_{i,t} | \mathcal{X}, y_{i,<t}, \tilde{\mathbf{h}}_i)}{\pi_{\text{ref}}(y_{i,t} | \mathcal{X}, y_{i,<t}, \tilde{\mathbf{h}}_i)}. \quad (8)$$

The unified training objective of LE-GRPO is formulated as:

$$\mathcal{L}_{\text{LE-GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \left[\min \left(r_{i,t}(\theta) \hat{A}_i, \text{clip}(r_{i,t}(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_i \right) - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right], \quad (9)$$

where $|y_i|$ is the sequence length of the i -th text response, and ϵ is the clipping hyperparameter. The group relative advantage $\hat{A}_i$ standardizes the multi-dimensional outcome reward $R(y_i)$ across the sampled group:

$$\hat{A}_i = \frac{R(y_i) - \text{mean}(R(y_1), \dots, R(y_G))}{\text{std}(R(y_1), \dots, R(y_G))}. \quad (10)$$

By fixing and replaying the continuous hidden states $\tilde{\mathbf{h}}_i$, we guarantee consistent conditional log-probabilities for the output text sequence, allowing the discrete outcome reward to seamlessly propagate backward and implicitly shape the upstream latent reasoning trajectory.

Multi-Dimensional Reward Design. The reward function $R(y_i)$ evaluates the holistic quality of the response by tracking three interconnected criteria: structural syntax, impurity control, and visual grounding accuracy.

Structural Syntax and Latent Indicator Reward (R_{struct}). We employ a regular expression validator to enforce both the macro response format and the activation of the latent workspace. A base reward is given if the output strictly conforms to the topology of $\langle \text{think} \rangle \dots \langle \text{/think} \rangle \langle \text{answer} \rangle \langle x_1 \rangle \langle y_1 \rangle \langle x_2 \rangle \langle y_2 \rangle \langle \text{/answer} \rangle$. Concurrently, to verify that the model explicitly triggers its inner reasoning process, an additional reward is granted if the special boundary markers $\langle \text{DAMA_START} \rangle$ and $\langle \text{DAMA_END} \rangle$ are successfully enclosed within the thinking block,

$$R_{\text{struct}}(y_i) = \begin{cases} 3.0 & \text{if format matched with boundary markers} \\ 2.0 & \text{if format matched without boundary markers} \\ -1.0 & \text{otherwise} \end{cases} \quad (11)$$

Impurity Penalty (R_{impurity}). To penalize structural drift or out of domain token leakage, an impurity penalty is stacked if a malformed response contains redundant HTML tags or parsing anomalies,

$$R_{\text{impurity}}(y_i) = \begin{cases} -1.0 & \text{if format failed with HTML tags} \\ 0.0 & \text{otherwise} \end{cases} \quad (12)$$

Visual Grounding Reward (R_{accuracy}). The core accuracy signal is determined by computing the Intersection over Union

(IoU) between the predicted bounding box B_{pred} parsed from the answer token and the ground truth bounding box B_{gt} . To ensure the model prioritizes robust formatting as the foundation for reliable grounding, the scaling weight of the IoU metric is dynamically configured based on syntax correctness,

$$R_{\text{accuracy}}(y_i) = \begin{cases} 5.0 \times \text{IoU}(B_{\text{pred}}, B_{\text{gt}}) & \text{if format matched} \\ 2.0 \times \text{IoU}(B_{\text{pred}}, B_{\text{gt}}) & \text{if format mismatched but box valid} \\ -3.0 & \text{if no box detected} \end{cases} \quad (13)$$

Total Unified Reward Score (R). The final reward score for each rollout is computed by aggregating the constituent reward components and is clipped within a strict numerical range of $[-5.0, 8.0]$ to mitigate extreme gradient variance.

$$R(y_i) = \max\left(-5.0, \min\left(R_{\text{struct}}(y_i) + R_{\text{impurity}}(y_i) + R_{\text{accuracy}}(y_i), 8.0\right)\right) \quad (14)$$

D. Hybrid Multi-LoRA Architecture

To mitigate high dimensional feature drift and potential gradient collapse caused by diverse degradation types, we propose a parameter-efficient mixture of degradation experts (MoDE) to enhance visual grounding performance. Specifically, our framework utilizes a lightweight gating module parameterized by a compact ViT architecture. Through this network, we execute a threshold-based dynamic soft fusion of specialized, degradation-specific LoRA experts within the high dimensional space. Consequently, the model can adaptively disentangle and process complex compound degradations in real world environments, which is formulated as follows,

$$G(V) = \text{Softmax}(W_g V), \quad \mathcal{A} = \{i \mid G(V)_i > \tau\} \quad (15)$$

where V denotes the high dimensional visual degradation embedding, W_g represents the learnable parameters of the lightweight gating network, and $G(V)_i$ signifies the routing weight assigned to the i -th expert. The dynamic subset $\mathcal{A}$ filters out irrelevant branches by isolating experts whose gating scores surpass the pre-defined activation threshold τ .

$$h_{\text{out}} = h_{\text{base}} + \sum_{i \in \mathcal{A}} \frac{G(V)_i}{\sum_{j \in \mathcal{A}} G(V)_j} \Delta W_i(h_{\text{base}}) \quad (16)$$

where h_{base} and h_{out} represent the input and output hidden states of the frozen transformer layer, respectively. The term $\Delta W_i(h_{\text{in}}) = B_i A_i h_{\text{in}}$ denotes the intrinsic forward computation of the i -th specialized LoRA expert, where A_i and B_i are the low rank down projection and up projection matrices.

E. Inference

During inference, the generation process of our DAMA approach unfolds through two distinct stages, latent visual reasoning and standard text generation for the final bounding box. Specifically, when the model generates the special marker $\langle \text{DAMA_START} \rangle$, it transitions into the latent reasoning mode to produce tokens that closely align with the visual semantics of the target object. Once this latent sequence concludes, the model switches back to the standard text generation mode to decode the final bounding box coordinates.

IV. EXPERIMENT

A. Datasets

To verify the effectiveness of our proposed approach, we conduct comprehensive experiments on two prominent RSVG datasets utilizing both quantitative analysis and qualitative evaluation.

- **DIOR-RSVG [50]** contains 17402 images sourced from the standard DIOR dataset [65], encompassing 38320 text queries and 20 distinct object categories. The dataset is partitioned into three subsets to facilitate systematic model training and evaluation. Specifically, the training set consists of 27133 text queries, the validation set comprises 2765 text queries, and the test set includes 7422 text queries.
- **VRSBench [30]** contains 29614 images and spans 26 object categories. Each image features one to five query instances, culminating in a total of 52472 text queries. This benchmark provides detailed descriptions of object attributes, including shape, position, color, relative position, and relative size. The dataset is divided into a training set encompassing 36313 text queries and a validation set containing 16159 text queries.

B. Experimental Setup and Baselines

To comprehensively evaluate the performance of the proposed approach, we employ multiple standard evaluation metrics, primarily focusing on intersection over union (IoU), which quantifies the spatial overlap between the predicted bounding boxes and the ground truth boxes. To ensure a rigorous and fair evaluation, we benchmark our proposed framework against a range of competitive baselines, including prominent vision-language models: GeoChat_{ft} [66], LLaVA-V1.5_{ft} [67], and Qwen-2.5-VL-3B_{ft} [69], as well as the specialized transformer-based approach, CSDNet [68]. For a consistent comparison, all baseline models are fine-tuned on the identical training dataset, as denoted by the subscript _{ft}.

All experiments are executed on a workstation equipped with four NVIDIA RTX 4090 GPUs. We adopt Qwen-2.5-VL-3B as our baseline architecture, which is fine tuned via LoRA with a rank of 32. The training framework utilizes DeepSpeed ZeRO stage 2 and leverages Bfloat16 mixed precision computation. During the supervised fine tuning stage, the model is trained on the VRSBench dataset for 26000 steps and the DIOR RSVG dataset for 16000 steps, both configured with a batch size of 16 and an initial learning rate of 2×10^{-4} . For the reinforcement learning stage, the rollout mechanism generates 24 samples per iteration with an optimized learning rate of 1×10^{-6} over a duration of 4000 steps. The trade off hyperparameter λ_{LVA} is set within the range of 0.08 to 0.14.

C. Comparison with SOTA approaches

On the VRSBench Benchmark, the quantitative evaluation results on the challenging VRSBench-Ref dataset across various multimodal disturbances are systematically compiled in Table I. The experimental data demonstrates that our proposed framework consistently achieves superior visual grounding

TABLE I: Visual grounding performance on the challenging VRSBench dataset across various image and text perturbations. Grounding accuracy is reported at an IoU threshold of 0.4. The subscript ft denotes models fine-tuned on the VRSBench training set.

Method	Venue	Naturalistic text perturbations					Persona text perturbations				
		S&P	Blur	Cloud	Compress	Avg.	S&P	Blur	Cloud	Compress	Avg.
GeoChat ft [66]	CVPR'24	14.31%	15.27%	15.55%	16.47%	15.40%	12.31%	13.55%	12.33%	9.67%	11.97%
LLaVA-V1.5 ft [67]	CVPR'24	11.84%	21.25%	14.99%	15.59%	15.92%	11.93%	12.64%	11.47%	12.14%	12.05%
CSDNet [68]	IF'24	32.25%	36.85%	33.56%	30.24%	33.23%	35.14%	39.58%	27.67%	32.63%	33.76%
Qwen-2.5-VL-3B ft [69]	Tech. Rep.	20.62%	19.82%	22.38%	17.80%	20.15%	23.10%	22.28%	24.36%	19.33%	22.27%
Ours (DA-SFT)	-	40.36%	36.81%	37.61%	27.61%	35.60%	44.51%	39.79%	40.04%	31.55%	38.97%
Ours (LE-GRPO)	-	41.06%	37.24%	39.09%	30.55%	36.99%	45.17%	40.21%	41.30%	33.05%	39.93%

TABLE II: Visual grounding performance on the challenging DIOR-RSVG dataset across various image and text perturbations. Grounding accuracy is reported at an IoU threshold of 0.4. The subscript ft denotes models fine-tuned on the DIOR training set.

Method	Venue	Naturalistic text perturbations					Persona text perturbations				
		S&P	Blur	Cloud	Compress	Avg.	S&P	Blur	Cloud	Compress	Avg.
GeoChat ft [66]	CVPR'24	32.16%	33.25%	30.19%	32.16%	31.94%	18.14%	18.54%	16.63%	17.50%	17.70%
LLaVA-V1.5 ft [67]	CVPR'24	25.86%	26.61%	24.89%	26.57%	25.98%	6.71%	6.94%	6.27%	6.93%	6.71%
CSDNet [68]	IF'24	30.87%	33.86%	29.88%	29.20%	30.95%	28.81%	31.31%	27.67%	27.22%	28.75%
Qwen-2.5-VL-3B ft [69]	Tech. Rep.	30.23%	29.35%	27.78%	32.61%	29.99%	28.63%	27.14%	25.42%	30.13%	27.83%
Ours (DA-SFT)	-	31.66%	33.67%	32.35%	33.98%	32.92%	30.63%	31.65%	29.78%	32.51%	31.14%
Ours (LE-GRPO)	-	32.85%	34.77%	32.83%	34.86%	33.83%	31.85%	32.77%	30.09%	33.35%	32.02%

accuracy compared to all competitive baselines under both naturalistic and persona text perturbations. Specifically, our LE-GRPO strategy establishes a new state of the art performance profile, achieving a 36.99% average grounding accuracy under naturalistic textual variations and a peak average of 39.93% under persona text formulations. These results represent a substantial margin of improvement over conventional MLLMs and specialized regression networks.

A granular analysis across individual degradation categories highlights the exceptional robustness of our architecture against severe environmental and sensor corruptions. Specifically, traditional models suffer severe performance drops under localized distortions and atmospheric occlusions, for instance, LLaVA-V1.5's accuracy plunges to approximately 11%–12% under salt-and-pepper noise, while the regression-based CSDNet plummets to 33.56% and 27.67% under cloud cover. **In sharp contrast**, our LE-GRPO-aligned model maintains highly robust performance landmarks, delivering 41.06% and 41.30% accuracy under natural and personalized textual perturbations, respectively. Furthermore, whereas standard vision language models like Qwen-2.5-VL-3B exhibit clear vulnerability when transitioning to complex conversational or emotional descriptions, hovering around 20.15% and 22.27% on average, our approach bridges this multimodal gap by seamlessly aligning the reasoning trajectory with stable visual anchors, making the framework immune to surface level query variations.

Furthermore, the transition from the DA-SFT phase to the reinforcement learning phase via LE-GRPO yields a consistent and universal performance boost across every single column in Table I. Specifically, the average grounding accuracy climbs

from 35.60% to 36.99% for naturalistic perturbations and from 38.97% to 39.93% for persona perturbations. This uniform advancement demonstrates that policy optimization through relative rewards does not merely overfit to specific corruptions, rather, it fundamentally optimizes the underlying latent representation space, unlocking the capability of the model to formulate more coherent and logical intermediate visual reasoning chains.

This performance gain is primarily attributed to our approach's emphasis on intrinsic latent space calibration, as opposed to conventional superficial data-level augmentation. In real world settings, visual corruptions like blur and noise trigger a severe high dimensional feature drift, causing representations to deviate from their standard clean distributions. This shift disrupts the precise text-visual alignment, forcing the model to operate on distorted representations and output inaccurate boundaries. Our manifold alignment paradigm addresses this limitation by projecting the shifted feature distributions back onto a stable, clean latent manifold. Anchoring the reasoning trajectory onto this corrected space successfully shields downstream semantic processing from environmental disturbances, ensuring robust visual grounding.

Additionally, we visualize the qualitative grounding performance on the VRSBench benchmark, as illustrated in Fig. 5. These visualizations clearly demonstrate that our method exhibits superior robustness under concurrent text and image perturbations. Even when the text instructions undergo naturalistic spelling corruptions, such as modifying 'large' to 'lauge', 'left' to 'lef', or 'white' to 'whiet', our approach maintains robust semantic reasoning capabilities. For example, in the second scenario containing the complex spatial phrase 'on the side

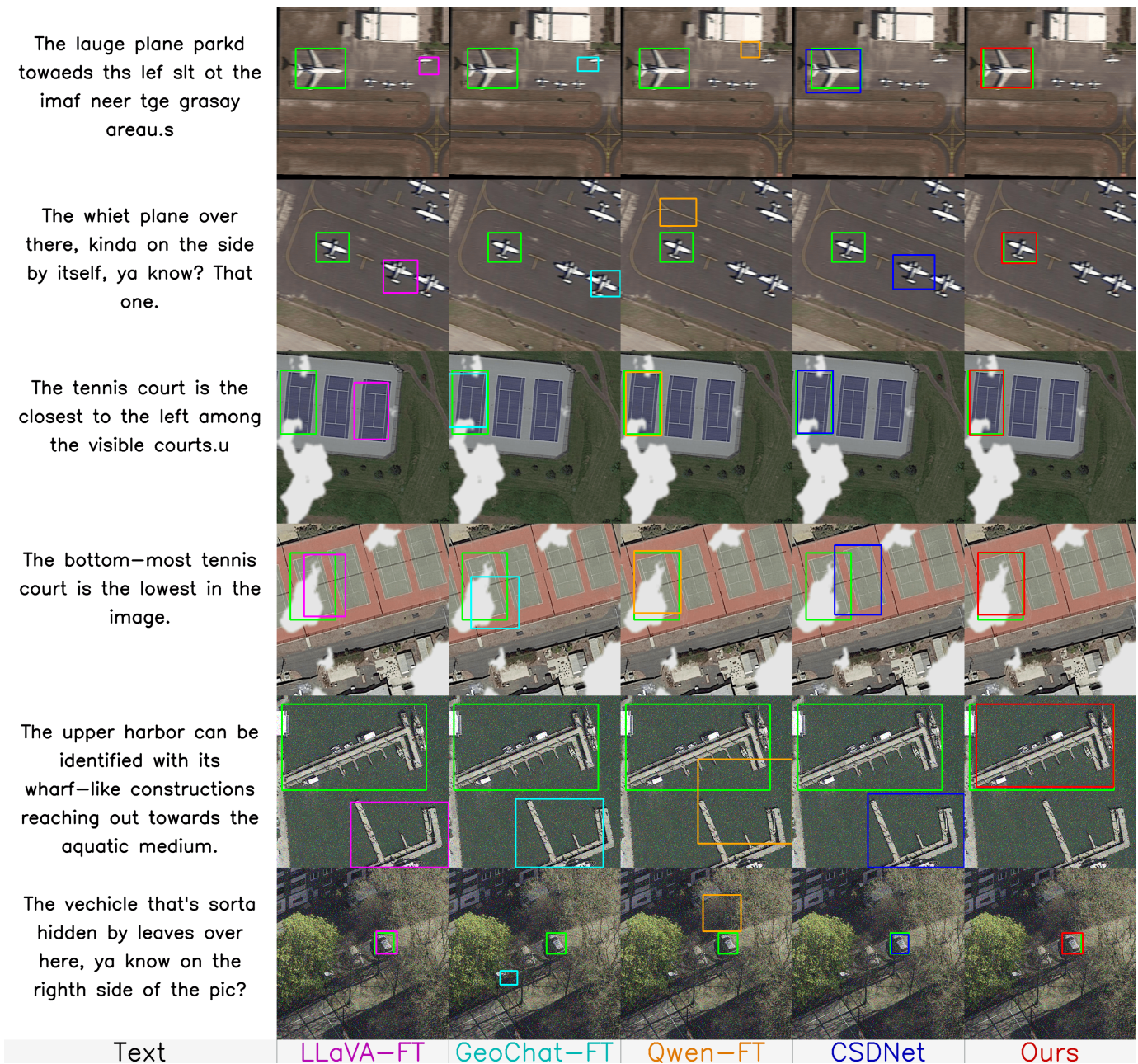


Fig. 5: Visualization of qualitative results comparing our proposed method against alternative approaches on the VRSBench benchmark. The green boxes represent the ground truth coordinates, whereas the other colored boxes indicate the predicted visual grounding boundaries. The input textual instructions are subject to naturalistic text perturbations. From top to bottom, the rows display image corruptions categorized into two cases of atmospheric blur, two cases of cloud cover, and two cases of salt and pepper noise.

by itself’, alternative approaches fail to capture the perturbed contextual cues and localize incorrect regions. In contrast, our model successfully interprets the underlying spatial layout, correctly identifying the isolated vehicle situated on the left side. This success highlights that our framework achieves significantly higher grounding accuracy by preserving stable reasoning trajectories under severe multimodal noise.

On the *DIOR-RSVG* Benchmark, the proposed approach acheive average 33.83%, and 32.02% on the naturalistic text prtrubations and persona text perturbation. This performance

gain show the proposed approach can improve grounding performance by employing dual anchored manifold alignment, which projecting the shifted feature distribution back on a stable, clean latent manifold, thus enabling robust performance on the text distructaion and image distructaion envirement. The visualize results are also shown in Fig. 6, the proposed approach generate bounding boxes that closely match the ground-truth annotations under degradation scenario, indicatig satisfactory localization performance.



Fig. 6: Visualization of qualitative results comparing our proposed method against alternative approaches on the DIOR RSVG benchmark. The green boxes represent the ground truth coordinates, whereas the other colored boxes indicate the predicted visual grounding boundaries. The input textual instructions are subject to naturalistic text perturbations.

D. Ablation Study

In this study, we perform comprehensive ablation studies on the VRSBench dataset to evaluate the effectiveness of our core contribution, DAMA alignment, and validate its potential value from various aspects.

Effectiveness of Dual-Anchored Manifold Alignment.

To investigate the individual contributions of the components within our DAMA framework, we conduct an ablation study isolating LVA and TSA. As reported in Table III, both LVA and TSA provide substantial improvements in visual grounding

accuracy under severe degradations. Specifically, LVA guides the model to reconstruct the clean visual features of target objects within the latent manifold. Concurrently, TSA prevents the internal representations from drifting into incoherent regions by enforcing strict cross-modal alignment with the expected textual ground-truth semantics. Driven by the synergistic effects of both anchors, our full framework achieves the optimal performance, outperforming the unaligned baseline by a massive margin (+19.74% under S&P noise and +16.99% under blur).

TABLE III: Ablation study of the proposed DAMA framework under Salt & Pepper (S&P) noise and blur perturbations. Accuracy (%) is reported to verify the effectiveness of Latent Visual Alignment (LVA) and Textual Space Alignment (TSA).

LVA	TSA	S&P Noise	Blur
✗	✗	20.62%	19.82%
✓	✗	32.22%	26.99%
✗	✓	31.04%	32.84%
✓	✓	40.36%	36.81%

Effectiveness of the Alignment Projector. We employ the alignment projector $\mathcal{P}_\theta$ to map the high-dimensional hidden states of the LLM into the visual semantic space of the target object. To evaluate its specific contribution, we conduct an ablation experiment by removing $\mathcal{P}_\theta$ while keeping all other architectural configurations identical. As reported in Table IV, omitting the alignment projector triggers a substantial performance drop, with the visual grounding accuracy declining sharply by 8.64% under S&P noise and 3.02% under blur. This precipitous degradation demonstrates the indispensable bridging role of $\mathcal{P}_\theta$ in cross-modal coordination. Without this calibrated mapping mechanism, directly injecting raw or mismatched visual semantics into the LLM tends to corrupt its latent representations, thereby disrupting downstream visual-semantic reasoning and deteriorating the final grounding accuracy.

TABLE IV: Ablation study investigating the impact of the alignment projector $\mathcal{P}_\theta$ under Salt & Pepper (S&P) noise and blur perturbations. Accuracy (%) is reported.

Projector $\mathcal{P}_\theta$	S&P Noise	Blur
✗	31.72%	33.79%
✓ (Ours)	40.36%	36.81%

Investigation of Visual Alignment Loss. We further investigate the impact of different loss functions for LVA by comparing mean squared error (MSE) and cosine similarity. As summarized in Table V, MSE yields superior performance across all perturbation types, achieving a remarkably large margin particularly under blur degradation. Crucially, we observe a severe decoding anomaly when employing cosine Similarity, the model’s internal reasoning process often fails to terminate, trapping the LLM in repetitive or endless generation loops. This phenomenon stems from the geometric properties of the loss functions. Cosine similarity solely enforces directional alignment while completely discarding the vector magnitudes of the hidden states. In LLMs, the norm of latent representations encapsulates vital semantic intensity and confidence calibration, which are directly tied to predicting termination tokens (e.g., `<|DAMA_end|>`). By scaling or distorting these norms arbitrarily, cosine alignment disrupts the model’s confidence thresholds, making it unable to conclude its thinking process. Conversely, MSE restricts both the orientation and the exact euclidean distance within the latent manifold, preserving the native scale of LLM representations and ensuring robust, well-bounded visual-semantic reasoning.

TABLE V: Comparison of different visual alignment loss functions under Salt & Pepper (S&P) noise and blur perturbations. Accuracy (%) is reported.

Loss Function	S&P Noise	Blur
Cosine Similarity	33.02%	15.85%
MSE (Ours)	40.36%	36.81%

Effectiveness of the Hybrid Multi-LoRA Architecture.

We investigate the architectural necessity of the proposed hybrid multi-LoRA design. Jointly optimizing a single adaptation network across highly disparate degradation categories often induces severe gradient conflicts, potentially leading to optimization collapse. To circumvent this issue, our framework configures dedicated, degradation-specific LoRA experts around a shared frozen backbone, thereby isolating the geometric adaptation paths for distinct corruptions. To verify its efficacy, we remove this multi-expert structure and instead conduct naive mixed training using a single standard LoRA variant. As demonstrated in Table VI, the naive mixed baseline suffers a catastrophic performance drop, collapsing to 34.90% under S&P noise and 31.84% under blur. This sharp degradation explicitly validates that joint training forces the model toward a compromised, suboptimal optimization direction due to conflicting gradient updates from distinct noise distributions. In contrast, our hybrid multi-LoRA architecture effectively decouples these interfering gradients, preserving distinct manifold correction trajectories. This architectural decoupling establishes a robust structural foundation, which we further optimize by employing on-policy distillation.

TABLE VI: Ablation study evaluating the hybrid multi-LoRA architecture under Salt & Pepper (S&P) noise and blur perturbations. Accuracy (%) is reported to demonstrate the impact of decoupling degradation-specific gradients.

Multi-LoRA Architecture	S&P Noise	Blur
✗ (Naive Mixed Training)	34.90%	31.84%
✓ (Ours)	40.36%	36.81%

V. CONCLUSION

In this paper, we presented DAMA, a robust framework designed to safeguard MLLMs against various image degradations and textual perturbations in visual grounding tasks. By integrating LVA and TSA, our method effectively constructs a robust latent space, preventing representation drifting and hallucinations during the intermediate reasoning process. To resolve gradient interference under multi-degradation settings, we developed a hybrid multi-LoRA architecture that successfully decouples the geometric correction trajectories. Furthermore, we leveraged LE-GRPO to eliminate over-regulation and optimize the internal reasoning flow through multi-dimensional outcome rewards. Extensive evaluations on the VRSBench and DIOR-RSVG benchmarks demonstrate that our framework consistently outperforms existing SOTA approaches by a substantial margin. Our research underscores the necessity of manifold alignment and architectural decoupling,

paving a promising avenue for deploying dependable MLLMs in real-world conditions.

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